

SpectraBayes: Exploring Traffic Series Reconstruction in Frequency Domain for Anomaly Detection

Ruoheng Li¹, Diyin Tang¹, *Member, IEEE*, Fei Wang², Xi Zhu¹, and Xianbin Cao¹, *Senior Member, IEEE*

Abstract—Timely detection of anomalies in traffic systems is crucial for mitigating risks and economic losses. Current time series anomaly detection methods often use reconstruction errors to identify anomalies, and their accuracy depends on how well they can reconstruct normal patterns from the original series. However, traffic data typically exhibit high noise, spatiotemporal heterogeneity, and uncertain inter-series correlations, complicating the learning of normal patterns. To address these challenges, we introduce SpectraBayes, which explores the reconstruction of density and volume series in the frequency domain for anomaly detection. First, we transform the series into the frequency domain and apply a low-pass filter to remove noise. Then, we embed periodic information into the frequency-domain representation through phase shifts to enhance the temporal awareness. Additionally, we model the inter-series correlations between density and volume resiliently using cross-spectrum probabilistic modeling. Optimized by maximizing the Evidence Lower Bound (ELBO), SpectraBayes ensures robust reconstruction while avoiding overfitting against the uncertain data. SpectraBayes outperforms 21 existing anomaly detection models on traffic series anomaly detection tasks, achieving mean improvements of 2.71% across three metrics over the second-best model. Furthermore, it is lightweight and maintains robust performance under varying noise levels.

Index Terms—Traffic anomaly detection, Bayesian neural network, frequency-domain representation.

I. INTRODUCTION

TRAFFIC anomalies, such as congestion or accidents, disrupt transportation systems and incur significant

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Ruoheng Li, Xi Zhu, and Xianbin Cao are with the School of Electronic and Information Engineering, Beihang University, Beijing 100191, China, also with the MIT Key Laboratory of Aerospace Mobile Communications, Beijing 100191, China, and also with the National Engineering Laboratory for Comprehensive Transportation Big Data Application Technology, Beijing 100191, China (e-mail: ruohengli@buaa.edu.cn; zhuxi@buaa.edu.cn; xbciao@buaa.edu.cn).

Diyin Tang is with the School of Automation Science and Electrical Engineering, Beihang University, Beijing 100191, China (e-mail: tangdiyin@buaa.edu.cn).

Fei Wang is with the State Key Laboratory of AI Safety, Institute of Computing Technology, Chinese Academy of Sciences, Beijing 100190, China, and also with the University of Chinese Academy of Sciences, Beijing 100049, China (e-mail: wangfei@ict.ac.cn).

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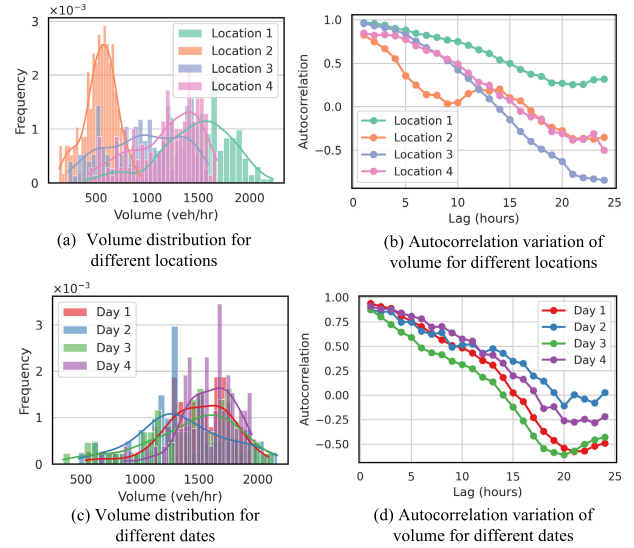


Fig. 1. Spatial-temporal heterogeneity of traffic volume, which is revealed by varied distribution and autocorrelation coefficient across different dates and locations. Data are selected from [1].

economic losses. Timely detection of such anomalies is crucial for traffic management and mitigation [1], [2].

Macroscopic traffic data, such as volume and density, are readily obtained from widely deployed loop detectors, providing a convenient resource for anomaly detection. Existing traffic time series anomaly detection methods can be broadly categorized as supervised or unsupervised. Supervised approaches rely on labeled data, limiting their generalization [3]. Early unsupervised methods assume normal patterns are in low-rank and detect anomalies as samples with large deviations [4], [5], [6]. However, these models often require large time windows for offline analysis. Recently, deep learning-based reconstruction methods enhance temporal modeling and anomaly identification by using the advanced architecture [7], [8].

However, traffic time series pose three main challenges for modeling normal patterns and detecting anomalies. First, they are highly noisy due to random individual behaviors and detector inaccuracies, which can impair model robustness and discrimination. Second, traffic exhibits pronounced spatiotemporal heterogeneity, as shown in Figure 1. Volume distributions and autocorrelation vary across locations and dates,

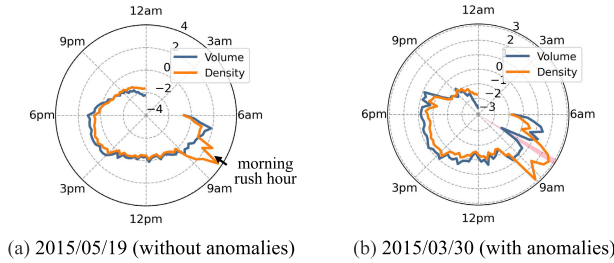


Fig. 2. Radar plots illustrate the normalized traffic volume and density across various dates at a specific location, revealing a variable correlation between these variables. Under normal conditions, the correlation between traffic volume and density varies within a range. A rapid negative correlation, marked by a red background, indicates potential anomalies. Data and labels are sourced from [1].

complicating the consistent definition of anomaly features. Third, although traffic flow and density are strongly correlated in ideal normal patterns [9], their relationship fluctuates under normal conditions and deviates during anomalies (Figure 2), highlighting the need for probabilistic modeling of inter-series correlations.

To address above challenging, we introduce SpectraBayes, a frequency-domain reconstruction model that utilizes the periodicity and cross-spectrum information for robust traffic series anomaly detection. Specifically, SpectraBayes employs the real Fast Fourier Transform (rFFT) for domain transformation and integrates a low-pass filter for data denoising. Then, SpectraBayes enhances the temporal awareness of model by embedding daily periodicity into the frequency domain. Meanwhile, the model explicitly quantifies correlations between traffic volume and density, leveraging the Bayesian neural parameters to adaptively learn the desirable correlated strength. The refined frequency representation is then reconstructed back to the time domain through an inverse FFT. The model is trained with the Evidence Lower Bound (ELBO) for probabilistic parameters estimation, which ensures both reconstruction accuracy and regularization.

The main contributions of this paper are as follows:

- 1) We propose SpectraBayes, to our knowledge, which is the first probabilistic frequency-domain reconstruction-based anomaly detection model that specifically designed for traffic series, which enables efficient temporal modeling and robust pattern reconstruction in an integrated framework.
- 2) We introduce a periodicity embedding mechanism that explicitly models daily cycles through phase-shifts in the frequency domain, enhancing detection performance by introducing temporal awareness.
- 3) We design a probabilistic cross-spectrum learning module that jointly models the spectral relationship between traffic density and volume, which can handle the aleatoric uncertainty inherent in traffic data.
- 4) Extensive evaluations against 21 state-of-the-art baselines confirm that SpectraBayes achieves superior detection performance (+2.71% over the second-best), along with significant improvements in computational efficiency and model robustness.

II. RELATED WORK

In this section, we review the current state of research on traffic time series anomaly detection, as well as the general time series representation models for anomaly detection.

A. Traffic Time Series Anomaly Detection Methods

Traffic time series anomaly detection methods can be broadly categorized into statistics-based, density-based, classification-based, and reconstruction-based approaches. Early statistical methods applied thresholds to macroscopic [10], [11], [12] and microscopic [13] variables, enabling initial anomaly identification but offering limited adaptability. Fundamental Diagram-based approaches [14], [15] detect deviations in flow-occupancy pairs, despite the strong interpretability, their reliance on manual calibration restricts robustness to dynamic traffic conditions. Incorporating nonparametric density-based methods, such as kernel density estimation [16] and FastABOD [1], can help identify abnormal flow-density pairs, but require retraining for different road segments.

Classification-based methods, including logistic regression [17], feedforward and recurrent neural networks [18], [19], and graph convolutional networks [20], learn mappings from spatiotemporal data to anomaly labels, but are limited by the availability of labeled data. Reconstruction-based approaches, encompassing tensor learning [5], [6], self-expressive models [21], and deep time-series representation models [7], [22], detect anomalies through deviations from learned normal patterns, offering finer temporal modeling but with performance varying across different setups [23]. Overall, traffic anomaly detection remains challenging due to system uncertainty and dynamics, motivating the development of tailored methods.

B. General Time Series Representation Models for Anomaly Detection

Time-series representation models can be broadly categorized into Encoder-Decoder and Encoder-only architectures. Classic Encoder-Decoder frameworks, such as Transformers [24], GANs [25], and VAEs [26], involve both encoding and decoding, increasing training complexity, whereas Encoder-only models offer simpler structures for series modeling [22], [27]. Recent studies suggest that merely increasing model complexity does not consistently improve anomaly detection [28], motivating approaches that explicitly capture anomaly-specific features, including inconsistency [29] and association discrepancies [7].

For traffic data, spatiotemporal heterogeneity, inherent daily periodicity, and stochastic variations due to road conditions and human factors complicate the modeling of general patterns. These characteristics should be explicitly considered when designing time-series representation models for traffic anomaly detection.

III. METHOD

A. Problem Formulation

Let $\{X_t^w\}_{t=1}^T$ denote the traffic time series within window w , where temporal vector at each time $X_t^w \in \mathbb{R}^2$ is

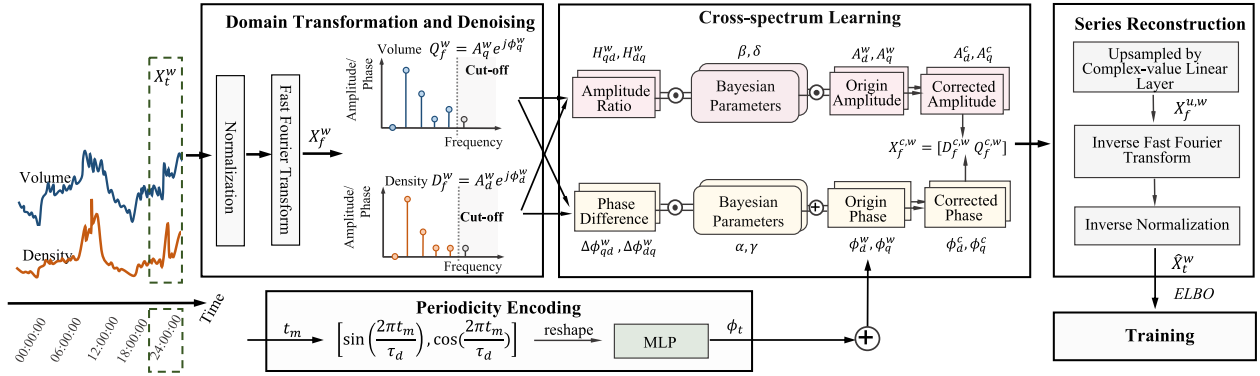


Fig. 3. Model structure of SpectraBayes. Input series X_t^w are normalized and transformed into the frequency domain as X_f^w using the real Fast Fourier Transform (rFFT), followed by low-pass filtering for denoising. Periodicity is then encoded through sinusoidal encoding as ϕ_t^w , which is embedded into frequency domain as phase shifts. To quantify the correlation between volume and density, amplitude ratios (H_{qd}^w, H_{dq}^w) and phase differences ($\Delta\phi_{qd}^w, \Delta\phi_{dq}^w$) are measured, with Bayesian parameters ($\alpha, \beta, \gamma, \delta$) adaptively adjusting their correlation strength. The resulted frequency representation is then upsampled using a complex-valued linear layer to obtain the $X_t^{u,w}$ with the required length for the inverse real Fast Fourier Transform (irFFT). The series then undergoes inverse normalization to recover the original scale. The Evidence Lower Bound (ELBO) optimizes the model, and reconstruction errors serve as anomaly indicators. During testing, anomaly scores are rated based on the reconstruction error $\|X_t^w - \hat{X}_t^w\|_2^2$.

defined as:

$$X_t^w = [Q_t^w \ D_t^w], \quad (1)$$

where Q_t^w and D_t^w represent traffic volume and density.

Our objective is to learn a reconstruction model f_ζ that reconstructs the normal traffic pattern from the original series $\hat{X}_t^w = f_\zeta(X_t^w)$. The anomaly scores is then determined by the reconstruction error:

$$s_t^w = \|\hat{X}_t^w - X_t^w\|_2^2. \quad (2)$$

An anomalous alert occurs when $s_t^w > \xi$, where ξ is a predetermined threshold.

B. Architecture

The SpectraBayes model, illustrated in Figure 3, first transforms each traffic time window $\{X_t^w\}_{t=1}^T \in \mathbb{R}^{T \times 2}$ into the spectral domain using rFFT and denoise it with a low-pass filter. Daily periodicity is embedded through sinusoidal phase shifts, and cross-spectral correlations between density and volume are refined through Bayesian learning. The spectrum is then inverted back to the time domain using irFFT, with model parameters optimized by maximizing the ELBO. During testing, the reconstruction error serves as the anomaly score. The details explanation for each component is presented in subsequent sections.

C. Domain Transformation and Denosing

A reversible instance-wise normalization (RIN) [22] is first applied on the time series X_t^w to mitigate the influence of a prominent zero-frequency component. The rFFT is then employed to convert the time-domain series X_t^w into the frequency domain, as expressed by:

$$X_f^w = \text{rFFT}(X_t^w) \in \mathbb{C}^{F \times 2}, \quad (3)$$

where $F = \lfloor T/2 \rfloor + 1$ denotes the number of frequency bins, with T denoting the size of detection window. X_f^w is a complex-valued matrix with dimensions $F \times 2$, including complex-valued elements for volume and density:

$$X^w(f) = [Q^w(f) \ D^w(f)]. \quad (4)$$

The complex value of volume and density $Q^w(f)$ and $D^w(f)$ can be expressed in polar form:

$$Q^w(f) = A_q^w(f)e^{j\phi_q^w(f)}, \quad D^w(f) = A_d^w(f)e^{j\phi_d^w(f)}, \quad (5)$$

where $A_q^w(f)$ and $A_d^w(f)$ represent the amplitudes of $Q^w(f)$ and $D^w(f)$, and $\phi_q^w(f)$ and $\phi_d^w(f)$ denote the phases of $Q^w(f)$ and $D^w(f)$.

To suppress noise in the frequency domain, a low-pass filter is applied on $X^w(f)$. Specifically, a cut-off frequency F_p is applied to retain frequency components below this threshold. The selection of F_p governs the denoising strength and can be determined by the harmonic content of the dominant frequency [22]. However, given the heterogeneity across datasets, we treat F_p as a hyperparameter, whose impact is explored in the parameter study.

D. Periodicity Encoding

Traffic series exhibit strong periodicity, and embedding periodicity into the data can enhance the capabilities of temporal modeling [30], [31]. In Transformer architectures, Sinusoidal Positional Encoding is commonly employed to provide positional awareness for the input. Drawing on the properties of the Fourier transform [32], a time-domain shift $x(t - \tau)$ corresponds to a phase shift in the frequency domain, as given by:

$$X_\tau(f) = A(f)e^{j(\phi(f) - 2\pi f\tau)}, \quad (6)$$

where τ denotes the time shift.

Inspired by this principle, we encode periodicity information as phase shifts to capture temporal features. Specifically, for a time window w containing T minute-of-day timestamps $t_m \in [0, 1439]$ ($m = 1, \dots, T$), we generate a base temporal embedding $e_d^w(t)$ for each timestamp following Time2Vec [30]:

$$e_d^w(t) = \left[\sin\left(\frac{2\pi t_m}{\tau_d}\right) \quad \cos\left(\frac{2\pi t_m}{\tau_d}\right) \right], \quad (7)$$

where $\tau_d = 1440$ minutes denotes the daily period.

Next, the base embeddings for the window are concatenated and flattened into a vector $e_d^w \in \mathbb{R}^{2T}$. The flattened temporal

embedding is then processed through a Multilayer Perceptron (MLP) and transformed into $\phi_t^w \in \mathbb{R}^{F_p}$ to ensure dimension alignment:

$$\phi_t^w = W_2 \cdot \sigma(W_1 e_d^w + b_1) + b_2, \quad (8)$$

where $W_1 \in \mathbb{R}^{D_t \times 2T}$ and $W_2 \in \mathbb{R}^{F_p \times D_t}$ are weight matrices, $b_1 \in \mathbb{R}^{D_t}$ and $b_2 \in \mathbb{R}^{F_p}$ are bias vectors, and σ is the ReLU activation function.

Finally, ϕ_t^w is applied to adjust the phases of both density ϕ_d^w and volume ϕ_q^w components, expressed as:

$$\tilde{\phi}_q^w = \phi_q^w + \phi_t^w, \quad \tilde{\phi}_d^w = \phi_d^w + \phi_t^w, \quad (9)$$

where $\tilde{\phi}_d^w$ and $\tilde{\phi}_q^w$ denote the adjusted phase.

E. Probabilistic Cross-Spectrum Learning

Incorporating inter-series information can enhance the reconstruction of normal patterns and improve anomaly detection performance, particularly in systems where variables exhibit strong correlations [33]. In the frequency domain, the relationship between volume (Q^w) and density (D^w) can be quantified using cross-spectral information: amplitude ratios and phase differences. For volume relative to density, the amplitude ratio $H_{qd}^w(f)$ and phase difference $\Delta\phi_{qd}^w(f)$ can be calculated as:

$$H_{qd}^w(f) = \frac{|Q^w(f)|}{|D^w(f)| + \epsilon}, \quad \Delta\phi_{qd}^w(f) = \arg\left(\frac{Q^w(f)}{D^w(f)}\right), \quad (10)$$

where ϵ is a small positive constant to prevent division by zero. Conversely, for density relative to volume:

$$H_{dq}^w(f) = \frac{|D^w(f)|}{|Q^w(f)| + \epsilon}, \quad \Delta\phi_{dq}^w(f) = \arg\left(\frac{D^w(f)}{Q^w(f)}\right). \quad (11)$$

The amplitude ratios reflect traffic efficiency, corresponding to the slope of the Fundamental Diagram under varying traffic states [9]. In ideal normal conditions, $H_{qd}^w(f)$ and $H_{dq}^w(f)$ approximate constants that determined by road characteristics. A high $H_{qd}^w(f)$ indicates free-flow pattern where volume dominates, while a high $H_{dq}^w(f)$ signals congestion where density prevails. Phase differences, meanwhile, encode propagation delays in traffic waves. Positive $\Delta\phi_{qd}^w(f)$ typically suggests downstream acceleration waves where volume changes precede density, whereas negative values indicate upstream shockwaves during congestion or incidents where density increases precede volume reductions.

Applying these cross-spectral information can be viewed as adaptive filtering between variables [34]. Parameters are introduced as learnable scalars to modulate the cross-correlation strength of cross-spectral learning. Considering the stochasticity of traffic data [35], [36], we employ Bayesian parameterization [37] within a variational inference framework to model cross-correlations strength probabilistically. Considering both of the computational efficiency and stochastic characteristic of traffic data [35], [36], we assign the priors of parameters as Gaussian distributions:

$$\alpha(f) \sim \mathcal{N}(\mu_\alpha(f), \sigma_\alpha^2(f)), \quad \beta(f) \sim \mathcal{N}(\mu_\beta(f), \sigma_\beta^2(f)), \quad (12)$$

$$\gamma(f) \sim \mathcal{N}(\mu_\gamma(f), \sigma_\gamma^2(f)), \quad \delta(f) \sim \mathcal{N}(\mu_\delta(f), \sigma_\delta^2(f)), \quad (13)$$

where μ and σ^2 denote the mean and variance of Gaussian priors. The Gaussian distribution enables closed-form updates and facilitates the optimization of the Evidence Lower Bound (ELBO) during training [37]. While other distributions can be used as alternatives, they may result in computational inefficiencies.

Based on the learnable parameters, the phase and amplitude of the spectra are then refined as:

$$\phi_d^{c,w}(f) = \tilde{\phi}_d^w(f) + (\Delta\phi_{qd}^w(f) \odot \alpha(f)), \quad (14)$$

$$A_d^{c,w}(f) = A_d^w(f) \odot (H_{qd}^w(f) \odot \beta(f)), \quad (15)$$

$$\phi_q^{c,w}(f) = \tilde{\phi}_q^w(f) + (\Delta\phi_{dq}^w(f) \odot \gamma(f)), \quad (16)$$

$$A_q^{c,w}(f) = A_q^w(f) \odot (H_{dq}^w(f) \odot \delta(f)), \quad (17)$$

where $\odot$ denotes element-wise multiplication.

The spectral representation that incorporates the cross-spectral information is denoted as:

$$X^{c,w}(f) = [Q^{c,w}(f) \ D^{c,w}(f)], \quad (18)$$

where:

$$Q^{c,w}(f) = A_q^{c,w}(f) e^{j\phi_q^{c,w}(f)}, \quad (19)$$

$$D^{c,w}(f) = A_d^{c,w}(f) e^{j\phi_d^{c,w}(f)}. \quad (20)$$

The refined spectra $X_f^{c,w}$ models and utilizes the dynamic correlation between traffic volume and density, allowing for a more integrated spectral representation.

F. Series Reconstruction and Uncertainty-Aware Training

Then, $X_f^{c,w}$ are upsampled through a complex-value linear layer:

$$X_f^u = W_c X_f^{c,w} + b_c, \quad (21)$$

where $W_c \in \mathbb{C}^{F \times F_p}$ and $b_c \in \mathbb{C}^F$ are learnable parameters of the complex-value linear layer, mapping the frequency bins from F_p back to F . After that, X_f^u is reconstructed back to time domain through the irFFT:

$$\hat{X}_t^w = \text{irFFT}(X_f^u). \quad (22)$$

Finally, the reconstructed series $\hat{X}_t^w$ undergoes an inverse RIN to reverse the initial normalization.

To train SpectraBayes, we employ variational inference to approximate the posterior $p(\theta|\mathcal{X})$ of the Bayesian parameters $\theta = \{\alpha, \beta, \gamma, \delta\}$ given the observed data $\mathcal{X}$. A Gaussian approximate posterior $q(\theta|\mathcal{X})$ is used to approximate $p(\theta|\mathcal{X})$, and the model is optimized by maximizing the Evidence Lower Bound (ELBO) [38]:

$$\mathcal{L} = \mathbb{E}_{q(\theta|\mathcal{X})}[\log p(\mathcal{X}|\theta)] - \text{KL}(q(\theta|\mathcal{X})||p(\theta)), \quad (23)$$

where $\mathbb{E}_{q(\theta|\mathcal{X})}[\log p(\mathcal{X}|\theta)]$ represents the expected log-likelihood of the observed data $\mathcal{X}$ with respect to the variational approximate posterior $q(\theta|\mathcal{X})$, encouraging the model to reconstruct the observed data. The second term in loss function $\text{KL}(q(\theta|\mathcal{X})||p(\theta))$ measures the Kullback-Leibler (KL) divergence between approximate posterior $q(\theta|\mathcal{X})$ and the prior $p(\theta)$ over all Bayesian parameters, acting as a regularizer. During training, θ is sampled using the reparameterization trick for gradient computation.

Assuming a Gaussian likelihood for the observed data, which can be given by $p(\mathcal{X}|\theta) = \prod_{i=1}^n \mathcal{N}(X_i|\hat{X}_i(\theta), \sigma_{\mathcal{X}}^2)$ (where i represents the i -th data point), the expected log-likelihood becomes:

$$\mathbb{E}_{q(\theta|\mathcal{X})}[\log p(\mathcal{X}|\theta)] = -\frac{n}{2} \log(2\pi\sigma_{\mathcal{X}}^2) - \frac{1}{2\sigma_{\mathcal{X}}^2} \sum_{i=1}^n \mathbb{E}_{q(\theta|\mathcal{X})}[(X_i - \hat{X}_i(\theta))^2]. \quad (24)$$

Based on the assumption, $\sigma_{\mathcal{X}}^2$ is a constant and can be omitted during optimization, the training objective is equivalent to minimizing:

$$\mathcal{L}_{\text{loss}} = \sum_{i=1}^n \mathbb{E}_{q(\theta|\mathcal{X})}[(X_i - \hat{X}_i(\theta))^2] + \text{KL}(q(\theta|\mathcal{X})||p(\theta)). \quad (25)$$

For Gaussian priors $p(\theta) \sim \mathcal{N}(\mu_p, \sigma_p^2)$ and posteriors $q(\theta|\mathcal{X}) \sim \mathcal{N}(\mu_q, \sigma_q^2)$, the KL divergence can be obtained by:

$$\text{KL}\left(\mathcal{N}(\mu_q, \sigma_q^2) \middle| \mathcal{N}(\mu_p, \sigma_p^2)\right) = \log\left(\frac{\sigma_p}{\sigma_q}\right) + \frac{\sigma_q^2 + (\mu_q - \mu_p)^2}{2\sigma_p^2} - \frac{1}{2}. \quad (26)$$

During test, means of the variational posterior $q(\theta|\mathcal{X})$ are used for inference, representing the most likely parameters.

G. Adaptive Anomaly Measurement

Reconstruction error s_t^w serves as the anomaly score for each detection window, reflecting deviations from learned traffic patterns. Anomaly is identified when s_t^w exceeds a threshold ϵ . Considering the spatiotemporal heterogeneity of traffic, we adopt two adaptive strategies to determine the threshold. First, a scenario-specific threshold is optimized on each validation dataset using grid search:

$$\epsilon_s = \arg \max_{\epsilon} \text{F1}(\epsilon; \mathcal{D}_{\text{val},s}). \quad (27)$$

Second, an adaptive threshold based on rolling statistics is employed, which adapts online during testing. Specifically, the threshold is updated within a sliding window of length W in a causal manner:

$$\epsilon_t(t) = \tilde{\mu}_t + k \cdot \tilde{\sigma}_t, \quad (28)$$

where $\tilde{\mu}_t$ and $\tilde{\sigma}_t$ denote the median and median absolute deviation (MAD) calculated from the window of size W that includes data from time $(t - W + 1)$ to time t . If t is less than W , the window will include all available data up to time t . The parameter k acts as a scaling factor, controlling the sensitivity of the threshold. It can be optimized offline on the validation data to accommodate diverse datasets.

H. Complexity Analysis

We analyze the computational complexity of SpectraBayes per detection window using big O notation [39], which provides asymptotic upper bounds while omitting constant factors and less significant contributions. Influencing parameters including the window size T , cutoff frequency F_p , and hidden dimension D_t for periodicity encoding. The time complexity

is dominated by rFFT ($O(T \log T)$), periodicity embedding ($O(D_t \times T)$), and frequency-domain reconstruction ($O(T \times F_p)$), resulting in an overall time complexity of

$$T(T, F_p, D_t) = O(\max(T \log T, D_t \times T, T \times F_p)). \quad (29)$$

The space complexity is mainly determined by storing temporal embeddings and frequency-domain representations, resulting in a space complexity of $S(T, F_p, D_t) = O(T \times D_t)$.

IV. EXPERIMENTS

We conducted various experiments to answer the following questions about our model: Q1. Accuracy. How accurately does our method detect anomalies compared to other baseline? Q2. Effectiveness. How do the modules designed in SpectraBayes work? Q3. Robustness. How does the performance vary under datasets of different noisy levels and parameter configurations?

A. Experimental Details

Below present details about our experimental setting, including dataset configurations, baselines for comparison, and parameter settings.

1) *Dataset Setup*: We adopt the traffic time-series anomaly dataset released by Sarteshnizi et al. [1], which consists of 10 sub-datasets sampled at 15-minute intervals. To obtain higher temporal resolution, we linearly interpolated the data to 1-minute intervals. Unlike real-world minute-level traffic measurements, which typically exhibit high stochasticity and low signal-to-noise ratios due to sensor noise and environmental factors [16], [40], interpolated series tend to be overly smooth. To better approximate realistic traffic fluctuations, we injected zero-mean Gaussian noise into the interpolated data.

The mean z-score of anomaly points in the original dataset is 0.27, indicating that excessive noise may introduce artificial anomalies and distort ground-truth labels. Accordingly, we set the noise variance to 0.01 to simulate mild stochasticity while preserving anomaly integrity. To further assess robustness, higher noise variances (0.04, 0.09, 0.16, and 0.25) were also evaluated, with results reported in Section IV-F.

The task is formulated as series-level anomaly detection. Time series are segmented into sequences using a sliding window with window size $T = 80$ and stride 1. Samples are split into training, validation, and testing sets with a 6:2:2 ratio. The average anomaly ratios across the 10 datasets are 26.36%, 20.12%, and 28.04%, respectively.

2) *Baselines*: To comprehensively evaluate the performance of SpectraBayes, we compared it with 21 representative anomaly detection methods covering density-based, distance-based, and reconstruction-based paradigms. Specifically, density-based methods include LOF [41], KDE [42], and ConFlow [43], which identify anomalies as low-density samples. ConFlow further models temporal dependencies through graph-augmented normalizing flows. Distance-based methods, including iForest [44], KNN [45], and Fast-ABOD [46], detect anomalies based on isolation characteristics or distance and angular deviations.

TABLE I
SELECTED BASELINES AND THE TUNED HYPERPARAMETERS

Model	Parameters	Search Space
LOF	n_neighbors	[3, 5, 8, 10, 15, 18, 20, 23]
iForest	n_estimators	[50, 100, 150, 200, 250]
KDE	bandwidth kernel	[0.5, 1, 1.5, 2, 2.5, 3] Gaussian
KNN	n_neighbors	[3, 5, 8, 10, 15, 18, 20, 23]
PCA	n_components	[0.6, 0.7, 0.8, 0.9, 0.95]
FastABOD	n_neighbors	[5, 8, 10, 13, 15, 18, 20, 23]
ConFlow	kernel_size flow_blocks	[5, 9, 13, 17, 21] [1, 2]
AE-LSTM	e_layer hidden_dim	[1, 2, 3] [32, 64, 128]
OmniAnomaly	nf_layers hidden_dim	2 [32, 64, 128]
MTAD-GAT	kernel_size hidden_dim	[7] [32, 64, 128]
Dlinear	kernel_size stride_size	[5, 9, 13, 17, 21] [1]
Transformer-based	e_layer	[1, 2, 3]
	model_dim attention_head	[32, 64, 128] [2, 4, 6, 8]
TimeNet	e_layer	[1, 2, 3]
	model_dim	[32, 64, 128]
	kernel	[3, 5, 7, 9]
FreTS	model_dim	[32, 64, 128]
FITS	Cut frequency	[13, 14, 15, 16, 17, 18]
SpectraBayes	Cut frequency	[13, 14, 15, 16, 17, 18]
	D_t	[32, 64, 128]
	μ_p	[0, 0.5, 1]
	σ_p	[0.1, 0.5, 1]

Note: Transformer-based models including InverseTransformer, PatchTST, Anomaly Transformer, DCdetector, TranAD, Freqformer and FEDformer. For PatchTST and DCdetector, number of patch size is search from [2, 4]. For FEDformer, the frequency mode is set as Wavelet.

Reconstruction-based methods include both classical and deep learning approaches. Classical PCA [47] identifies anomalies by low-rank reconstruction errors. Deep learning-based models span recurrent architectures (AE-LSTM, OmniAnomaly [8]), graph neural networks (MTAD-GAT [33]), convolutional models (TimesNet [48]), linear models (Dlinear [49], FreTS [50], FITS [22]), and Transformer-based methods, including Anomaly Transformer [7], TranAD [24], DCdetector [29], InverseTransformer [51], PatchTST [52], FEDformer [53], and Freqformer [54]. These models enhance temporal representation ability of Transformer architecture through attention mechanisms, patch-wise modeling, adversarial training, dual-attention designs, or frequency-domain modeling.

3) *Experimental Setting:* All experiments were conducted on a server running Ubuntu 20.04.5 LTS, equipped with 192 GB of RAM, an Intel Xeon Gold 6242R CPU @ 3.10 GHz, and an NVIDIA GeForce RTX 4090 GPU. All models were implemented using PyTorch version 1.7.1 with

CUDA 12.4. Each model was trained for 40 epochs using the Adam optimizer, with a learning rate of 0.0001 and a weight decay of 0.0005. Optimal hyperparameters for each model were determined through a parameter search, with candidate values detailed in Table I. For Transformer-based architecture and TimeNet, a fixed dropout rate of 0.1 and the GELU activation function were applied. All models are trained and evaluated five times with fixed random seeds, with mean performance reported. Deterministic methods, such as LOF, KDE, Fast-ABOD, KNN, and PCA, had identical results across runs.

4) *Evaluation Metrics:* Model performance is evaluated using the Area Under the Receiver Operating Characteristic Curve (AUC-ROC) and the F1 score (F1). AUC-ROC measures the trade-off between true positive and false positive rates across varying thresholds, reflecting the overall discriminative capability. The F1 score evaluates detection performance at a specific decision threshold.

To account for spatiotemporal heterogeneity, we adopt both offline scenario-specific thresholding and an online dynamic thresholding strategy, which is explained in detail in Section III-G. The offline threshold is optimized for using grid search on validation data. The online threshold is determined through a rolling statistics method that adapts causally during testing. For online thresholding, the window size is fixed to $W = 500$ for all methods, while the scaling factor $k \in \{0.5, 1, 1.5, 2, 2.5, 3\}$ is selected based on validation performance.

B. Comparative Study

We compared SpectraBayes with 21 baseline models using three evaluation metrics: AUC-ROC (Table II), F1-score with a scenario-specific threshold ϵ_s (F1- ϵ_s , Table III), and F1-score with a rolling-statistic threshold ϵ_r (F1- ϵ_r , Table IV). Generally, the scenario-specific threshold ϵ_s , determined through a fine-grained search on validation datasets, achieved higher detection accuracy than the rolling-statistic threshold ϵ_r . Across all datasets and metrics, SpectraBayes demonstrated superior performance, achieving the best results on 9 out of 10 datasets in terms of AUC-ROC, 7 out of 10 for F1- ϵ_s , and 8 out of 10 for F1- ϵ_r . Notably, SpectraBayes outperformed the second-best method by up to 4.65% in AUC-ROC (dataset 8-E), 6.21% in F1- ϵ_s (dataset 1-W), and 3.03% in F1- ϵ_r (dataset 8-E). On average, SpectraBayes consistently led across all three metrics, surpassing competitors such as InverseTransformer and PatchTST by ranges from 1.78% to 3.3%.

Classical machine learning methods, including LOF, iForest, KNN, FastABOD, PCA, and KDE, exhibited limited effectiveness due to their inability to model temporal patterns. Among these, PCA achieved the highest average AUC-ROC (61.08%), KNN led in F1- ϵ_s (46.73%), and PCA led in F1- ϵ_r (38.54%). ConFlow, which employs conditional normalizing flows to integrate temporal information, outperformed traditional methods with an average AUC-ROC of 74.02%, F1- ϵ_s of 50.37%, and F1- ϵ_r of 47.04%. However, its performance was constrained by the noisy nature of traffic data, which diminished the discriminative ability of density-based approaches.

TABLE II
COMPARISON OF AUC-ROC (%) BETWEEN SPECTRABAYES AND BASELINE MODELS

Model \ Dataset	Melbourne						Seattle				Average
	1-N	1-W	8-E	14-E	21-W	29-S	d005	d090	i005	i090	
LOF	51.38	51.60	50.22	50.56	59.40	44.11	41.73	55.28	67.57	49.89	53.17
iForest	53.11	44.81	58.26	56.42	43.68	60.07	62.36	50.40	67.05	56.34	55.25
KNN	52.99	53.20	75.74	62.67	41.53	79.18	71.21	50.78	61.42	55.91	60.46
FastABOD	52.11	51.38	72.62	61.3	41.0	77.44	70.65	48.65	46.48	55.46	57.61
PCA	58.43	52.92	64.84	60.49	54.21	69.69	66.39	53.41	68.17	62.21	61.08
KDE	58.08	47.78	60.58	57.89	52.26	64.17	64.64	48.36	62.78	56.36	57.29
ConFlow	69.54	80.91	<u>83.20</u>	73.55	70.49	80.02	71.17	62.46	<u>83.18</u>	65.63	74.02
AE-LSTM	69.16	65.72	77.70	71.10	55.51	81.69	66.06	58.77	75.12	64.26	68.51
OmniAnomaly	62.94	66.79	75.62	66.52	59.00	78.09	66.22	63.02	71.82	63.70	67.37
MTAD-GAT	54.24	56.04	68.84	61.75	53.43	69.51	65.37	59.67	60.85	67.77	61.75
AnomalyTransformer	56.17	60.53	65.46	59.81	55.81	59.44	61.94	46.89	55.56	52.76	57.44
DCdetector	52.12	55.90	67.20	65.35	58.70	61.18	69.42	65.38	74.29	54.41	62.40
TranAD	54.70	57.05	58.28	56.03	54.41	61.62	65.37	47.88	54.28	60.29	56.99
TimeNet	69.68	76.53	74.36	69.79	70.87	80.07	69.38	67.15	75.07	62.06	70.50
FEDformer	62.67	56.42	65.44	61.86	56.08	68.93	66.31	53.51	69.03	57.62	61.79
Freqformer	72.35	74.33	81.80	71.87	56.71	81.94	68.50	60.01	77.28	64.87	70.97
PatchTST	76.33	82.00	81.50	72.69	74.71	81.72	72.04	71.21	76.11	68.48	75.68
InverseTransformer	76.40	82.30	80.44	<u>74.10</u>	76.49	<u>82.04</u>	<u>72.79</u>	71.49	78.97	68.76	<u>76.38</u>
Dlinear	<u>77.57</u>	<u>82.49</u>	82.17	73.16	74.19	81.36	71.64	<u>71.63</u>	76.82	69.48	76.05
FreTS	76.25	81.56	81.6	71.15	70.61	80.53	71.54	<u>71.33</u>	75.82	<u>69.62</u>	75.00
FITS	73.73	77.47	78.51	71.30	72.82	81.99	70.77	69.75	74.66	66.81	72.88
SpectraBayes	78.41	85.79	86.82	75.27	<u>75.99</u>	82.89	73.80	71.96	83.41	70.08	79.44

Note: The AUC-ROC(%) is reported as mean of five random seeds. The best and second-best results are marked in **bold** and with underline.

TABLE III
COMPARISON OF F1-SCORE (%) BETWEEN SPECTRABAYES AND BASELINE MODELS WITH SCENARIO-SPECIFIC THRESHOLD ϵ_s

Model \ Dataset	Melbourne						Seattle				Average
	1-N	1-W	8-E	14-E	21-W	29-S	d005	d090	i005	i090	
LOF	23.75	30.04	13.05	47.15	23.97	25.24	29.94	15.89	14.00	15.22	23.83
iForest	48.99	26.93	38.76	41.21	42.48	33.52	18.83	28.19	29.52	25.39	33.38
KNN	63.79	36.37	58.92	51.08	43.69	59.94	38.61	28.35	53.01	33.53	46.73
FastABOD	63.86	36.76	56.43	50.8	43.47	58.64	38.2	27.99	52.02	33.61	46.18
PCA	63.64	35.19	50.85	52.91	44.86	55.5	41.43	20.57	55.81	36.78	45.75
KDE	62.88	38.00	47.90	48.68	44.19	51.28	31.96	29.02	51.78	31.64	43.73
ConFlow	63.78	54.71	65.61	51.12	48.88	52.41	32.16	35.09	63.08	36.90	50.37
AE-LSTM	62.04	39.10	62.25	56.33	44.43	61.09	37.95	24.12	56.44	37.74	48.15
OmniAnomaly	60.03	40.23	54.94	51.85	42.16	56.95	32.75	35.7	54.29	37.41	46.63
MTAD-GAT	63.65	40.08	51.46	50.75	44.18	51.72	33.66	31.32	51.41	38.48	45.67
AnomalyTransformer	63.58	37.71	47.98	59.46	42.93	45.13	27.44	29.76	48.04	33.34	43.54
DCdetector	62.86	37.47	46.62	50.90	42.57	45.59	27.23	29.71	49.64	32.61	42.52
TranAD	63.63	37.78	48.01	50.75	44.14	43.89	32.24	23.07	45.72	33.78	42.30
TimeNet	68.35	56.69	62.25	57.63	48.13	65.88	37.52	37.58	61.96	39.38	53.54
FEDformer	65.17	33.71	49.70	51.11	44.22	53.74	32.07	30.24	55.08	34.74	44.98
Freqformer	67.96	47.08	58.3	<u>60.46</u>	48.31	60.61	35.51	37.39	64.11	42.68	52.24
PatchTST	64.74	59.33	60.96	<u>60.25</u>	55.11	<u>67.51</u>	37.42	41.47	64.73	41.06	55.26
InverseTransformer	65.70	59.04	64.45	58.81	54.32	65.25	39.40	42.63	63.28	41.53	<u>55.44</u>
Dlinear	65.78	<u>59.91</u>	<u>65.25</u>	59.39	50.58	66.13	38.46	39.67	63.98	41.62	55.08
FreTS	<u>69.00</u>	54.90	61.57	59.53	49.52	59.21	<u>41.52</u>	44.33	<u>65.36</u>	43.87	54.88
FITS	63.86	52.64	62.29	58.47	<u>54.93</u>	66.52	37.42	39.11	63.48	40.34	53.91
SpectraBayes	70.51	66.12	69.83	61.07	54.06	68.18	42.98	<u>43.23</u>	68.60	<u>42.84</u>	58.74

Note: The results are reported as mean of five random seeds. The best and second-best results are marked in **bold** and with underline, respectively.

Among deep learning models, simpler linear architectures often outperformed more complex ones. SpectraBayes, Dlinear, FITS, and FreTS are linear-based models that operate in the time or frequency domain, performing better than complex architectures specifically designed for multivariate time series. Dlinear, which decomposes series into high- and low-frequency components for separate temporal modeling, achieved the third-highest average AUC-ROC (76.05%). Inter-series correlation remains important for improving detection

accuracy. FreTS, with its Frequency Channel Learner for modeling inter-series correlations, outperformed FITS by 2.12% in AUC-ROC and by 0.97% in F1- ϵ_s . Complex architectures designed for multivariate time series, such as MTAD-GAT (which uses a dual graph attention network for fine-grained inter- and intra-series correlation modeling), exhibited unsatisfactory performance (AUC-ROC: 61.75%, F1- ϵ_s : 45.67%, F1- ϵ_t : 37.22%). OmniAnomaly performed better than MTAD-GAT by leveraging stochastic neural networks (AUC-ROC:

TABLE IV
COMPARISON OF F1-SCORE (%) BETWEEN SPECTRABAYES AND BASELINE MODELS WITH ROLLING-STATISTIC THRESHOLD ϵ_r

Model \ Dataset	Melbourne						Seattle				Average
	1-N	1-W	8-E	14-E	21-W	29-S	d005	d090	i005	i090	
LOF	23.75	30.04	13.05	7.15	23.97	25.24	29.94	15.89	14.00	15.22	19.83
iForest	47.52	30.55	37.94	39.24	30.10	42.45	35.49	20.89	55.77	32.62	37.26
KNN	35.12	32.84	58.38	48.86	17.87	43.33	31.87	19.33	37.11	16.24	34.10
FastABOD	41.72	24.07	43.78	40.21	22.29	45.4	32.49	21.23	26.03	29.75	32.70
PCA	52.02	30.34	43.58	46.54	28.74	41.66	35.26	28.61	42.04	36.56	38.54
KDE	44.64	28.66	37.76	44.86	34.15	43.58	33.33	22.22	46.88	33.01	36.91
ConFlow	62.90	41.71	56.80	50.68	46.64	53.12	32.28	31.45	59.74	35.11	47.04
AE-LSTM	52.82	33.60	53.54	48.42	28.94	53.32	35.98	29.34	59.46	37.91	43.33
OmniAnomaly	52.89	38.82	52.69	49.78	37.53	51.52	32.20	34.72	52.29	37.04	43.95
MTAD-GAT	47.24	28.09	43.72	43.56	27.87	42.74	31.37	29.32	40.17	38.14	37.22
AnomalyTransformer	44.45	25.44	26.67	54.68	11.55	19.9	16.83	17.07	42.94	21.47	28.10
DCdetector	57.96	28.35	35.57	43.30	39.16	39.33	25.18	28.12	40.08	23.60	36.07
TranAD	44.99	29.54	40.01	41.30	36.31	37.69	31.15	20.36	39.06	31.90	35.23
TimeNet	57.66	49.02	57.22	54.70	50.91	55.70	37.16	37.38	57.25	39.52	49.65
Freqformer	61.47	40.46	55.56	51.71	31.88	57.62	35.43	30.09	<u>60.53</u>	36.38	46.11
FEDformer	52.97	29.20	42.33	46.46	37.89	37.67	35.07	27.50	<u>49.42</u>	29.90	38.84
PatchTST	62.18	51.04	60.22	<u>55.21</u>	<u>53.88</u>	<u>59.37</u>	37.89	39.39	56.28	41.24	<u>51.67</u>
InverseTransformer	<u>62.70</u>	48.42	<u>60.32</u>	54.29	53.12	58.62	38.67	39.15	52.61	42.44	51.03
Dlinear	58.44	47.94	58.41	52.77	50.96	55.85	35.07	<u>39.58</u>	56.64	39.88	49.55
FreTS	58.72	47.98	57.31	50.77	48.56	55.52	34.97	39.35	55.99	<u>42.26</u>	49.14
FITS	58.70	44.46	57.17	55.09	53.36	57.84	36.30	38.18	53.69	42.17	49.70
SpectraBayes	64.40	52.33	63.35	55.63	53.92	62.00	<u>38.02</u>	40.27	62.69	41.90	53.45

Note: The results are reported as mean of five random seeds. The best and second-best results are marked in **bold** and with underline, respectively.

67.37%, $F1-\epsilon_s$: 46.63%, $F1-\epsilon_r$: 43.95%). However, it did not explicitly incorporate inter-series correlations, resulting in limited performance similar to that of AE-LSTM. The convolutional variant, TimeNet, achieved superior performance (AUC-ROC: 70.50%, $F1-\epsilon_s$: 53.54%, $F1-\epsilon_r$: 53.54%) compared to OmniAnomaly, owing to its use of TimeBlocks for embedding multi-periodicity.

Transformer-based methods exhibited varied performance. InverseTransformer, which tokenized entire series for global correlation modeling, achieved second-best results in AUC-ROC (76.38%) and $F1-\epsilon_s$ (55.44%). PatchTST, employing a patching strategy for patch-wise dependencies, secured the second-best $F1-\epsilon_r$ (51.67%). However, the frequency-variant Transformer architecture FEDformer performed poorly due to its embedded random frequency sampling mechanism, which retained many noisy components (AUC-ROC: 61.79%, $F1-\epsilon_s$: 44.98%, $F1-\epsilon_r$: 38.84%). Freqformer, a frequency-domain Transformer without such sampling, performed better (AUC-ROC: 70.97%, $F1-\epsilon_s$: 52.24%, $F1-\epsilon_r$: 46.11%). Transformer-based models that specifically designed for time series anomaly detection tasks (AnomalyTransformer, DCdetector, and TranAD) exhibited suboptimal performance on noisy traffic data, as their discriminative strategies, such as association or representation discrepancy, were less effective with uncertain data.

These results highlight that complex architectures do not guarantee superior performance. Instead, decoupled mechanisms (Dlinear), inter-series correlation modeling (InverseTransformer, PatchTST), and frequency-domain representations (FreTS, Freqformer, SpectraBayes) enhance robustness against noisy traffic data. SpectraBayes, due to its effective denoising and probabilistic correlation modeling, consistently achieving the best performance.

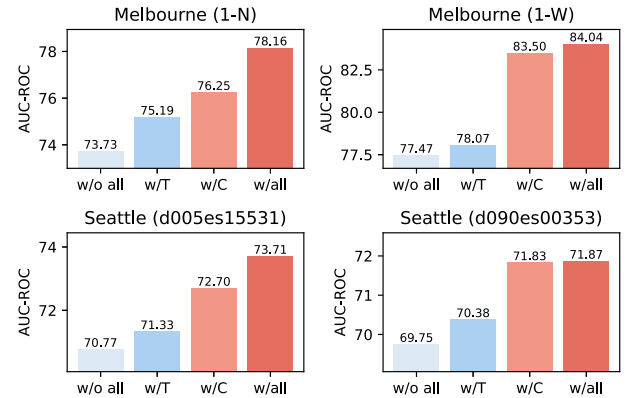


Fig. 4. Ablation Results. Baseline (w/o all), with periodicity encoding modules (w/ T), with cross-spectrum modules (w/ C), full model (w/ all).

C. Ablation Study

We evaluated the effectiveness of the modules designed in SpectraBayes by testing various combinations of modules, including temporal embedding (T), and cross-spectrum learning (C) module. The results of ablation study are illustrated in Figure 4.

The baseline model (w/o all) demonstrated varied performance across datasets, ranging from 69.75% to 77.47%. When evaluating the individual contributions of each module, the incorporation of Periodicity Encoding (w/ T) and cross-spectrum learning (w/ C) module consistently improved performance across all datasets. Specifically, the inclusion of module C alone led to more significant performance gains (2.52% on 1-N dataset, 6.03% on 1-W dataset, 1.93% on d005es1553 dataset, 2.08% on d090es0035 dataset). In

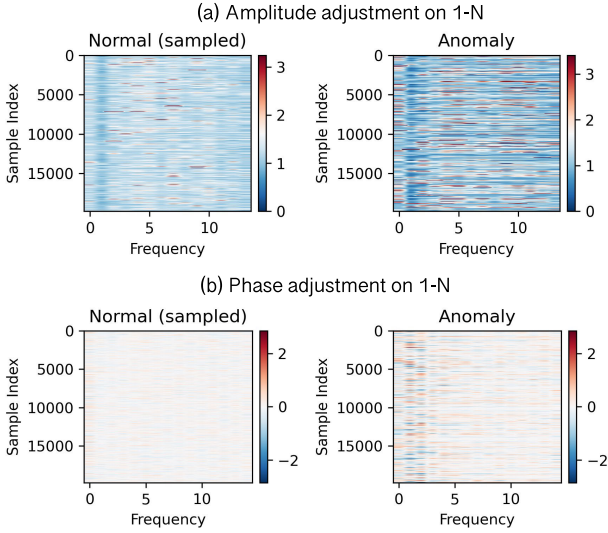


Fig. 5. Visualization of amplitude and phase adjustment for 1-N datasets. Normal series are sampled to keep the same number with anomalous series.

contrast, Periodicity Encoding module achieves smaller performance gains (1.46% on 1-N dataset, 0.6 % on 1-W dataset, 0.56% on d005es1553 dataset, 0.63% on d090es0035 dataset). When both modules were integrated (w/ all), the model achieved the best detection performance across all datasets. Notably, on Melbourne subset 1-W, the combination of all modules boosted performance to 84.68%, representing an improvement of 6.57% over the baseline model.

D. Module Analysis

1) *Visualization of Cross-Spectrum Learning*: To demonstrate the impact of our cross-spectrum module, we visualized the amplitude and phase adjustments for density in cross-spectrum learning, as shown in Figure 5. These heatmap showed obvious difference between normal and anomalous series. For instance, in Figure 5 (a), amplitude adjustments for the 1-N dataset show that normal samples exhibit uniform and minimal changes, consistently near 1.0, indicating negligible modification of the original signal. In contrast, anomalous samples display adjustments spanning a wider range (from 0 to 3), reflecting significant amplitude discrepancies between density and volume. Regarding phase adjustments, normal samples show minimal and uniform shifts, clustering near zero, as depicted in Figure 5 (b). However, anomalous samples, particularly in the low-frequency range, exhibit larger phase shifts ($-\pi$ to π). By leveraging cross-spectrum learning, SpectraBayes enhances the discriminability of reconstructed sequences in the feature space, thereby improving detection performance.

2) *Effectiveness of the Bayesian Parameters*: To assess whether incorporating probabilistic modeling can enhance detection robustness, we compared two model variants across datasets with varying noise intensities, where Gaussian noise variance was set to different levels. The first variant (non-probabilistic) adopted conventional deterministic parameters in cross-spectrum learning optimized through stochastic gradient

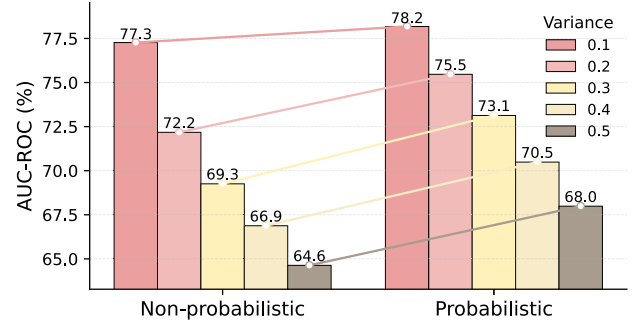


Fig. 6. Performance comparison between probabilistic and non-probabilistic learning under various noisy level.

descent, which is a frequentist point estimate. In contrast, the second (probabilistic) employed a Bayesian formulation that explicitly modeled aleatoric uncertainty through probabilistic parameter estimation. As illustrated in Figure 6, the Bayesian model consistently outperformed its deterministic counterpart across all noise levels, achieving up to a 3.8% improvement in AUC-ROC. This consistent gain demonstrates the benefit of uncertainty-aware modeling, which enables the system to better capture the stochasticity inherent in traffic data and maintain better performance under noisy conditions.

E. Computational Efficiency Comparison

Computational efficiency comparison experiments were conducted on an Ubuntu 20.04.5 LTS server equipped with an Intel Xeon Gold 6242R CPU (192 GB RAM) and an NVIDIA RTX 4090 GPU. We evaluated the computational efficiency and deployment feasibility of SpectraBayes compared to the baselines (TimesNet, AnomalyTransformer, DCdetector, PatchTST, InverseTransformer, FEDformer, and FreTS). The metrics evaluated are the number of trainable parameters, Multiply-Accumulate Operations (MACs), and average inference time per detection window, which correspond to the space complexity, theoretical computational complexity, and empirical computational complexity, respectively. The low number of trainable parameters and MACs suggests minimal memory usage and computational demand on resource-constrained hardware, while the short inference time demonstrates the high computational efficiency. All measurements were obtained using the standard PyTorch profiler, with inference times averaged over five runs. For Transformer-based models (AnomalyTransformer, DCdetector, PatchTST, InverseTransformer, and FEDformer), the number of layers, attention heads, and hidden dimensions were set to 2, 4, and 128, respectively. For the convolutional model (TimesNet), the layer depth, kernel size, and hidden dimension were fixed at 2, 3, and 128. The frequency-domain model (FreTS) used a hidden dimension of 128. For SpectraBayes, the cut-frequency F_q and temporal embedding dimension were set to 15 and 128.

The results in Table V show that SpectraBayes achieved the lowest values across all metrics, demonstrating superior efficiency. It required only 0.08 M trainable parameters, approximately 38.5% fewer than the second-lightest model (DCdetector, 0.13 M). In terms of theoretical computation,

TABLE V

COMPARISON OF MODEL PARAMETERS, MACs, AND INFERENCE TIME FOR THE 1-N DATASET WITH A DETECTION WINDOW SIZE OF $T = 80$

Model	Parameters	MACs	Inference Time
TimesNet	2.30 M	550.67 M	0.140ms
AnomalyTransformer	0.20 M	16.10 M	0.033ms
DCdetector	0.13 M	19.30 M	0.018ms
PatchTST	0.50 M	8.16 M	0.013ms
InverseTransformer	0.42 M	<u>1.67 M</u>	<u>0.008ms</u>
FEDformer	0.73 M	183.93 M	1.081ms
FreTS	2.64 M	5.33 M	0.012ms
SpectraBayes (Ours)	0.08 M	0.13 M	0.002ms

Note: MACs denote Multiply-Accumulate operations, measuring computational complexity through multiplication and addition operations during inference. The best and second-best results are in **bold** and underlined, respectively.

SpectraBayes used just 0.13 M MACs, over ten times fewer than the next-best baseline (InverseTransformer, 1.67 M) and far below larger models like TimesNet (550.67 M). The empirical inference time per window for SpectraBayes was 0.002 ms, which was four times faster than the second-fastest baseline (InverseTransformer, 0.008 ms) and over 500 times faster than the slowest (FEDformer, 1.081 ms).

F. Robustness Analysis

1) Robustness Comparison Under Different Noise Levels:

To evaluate the model robustness under varying noise conditions, we conducted comparative experiments on datasets with different levels of injected Gaussian noise. We chose Dlinear, FreTS, InverseTransformer, and PatchTST for comparison since they achieved competitive performance in the comparative study under low noise levels. Specifically, the noise level was parameterized by the variance of the Gaussian noise, ranging from 0.01 to 0.25 times the variance of the original data.

Results are shown in Figure 7. Specifically, increasing noise levels led to performance degradation for all models, highlighting the challenges of increased data uncertainty. SpectraBayes consistently outperformed the baselines at every noise level, achieving the highest AUC-ROC scores consistently. With the increasing of the noisy level, its performance had an absolute drop of 11.4% and a relative decline of 14.36%. In comparison, Dlinear showed the largest degradation, with scores dropping from 76.1% to 60.4% (absolute drop of 15.7%, relative drop of 20.63%). FreTS, PatchTST, and InverseTransformer exhibited relative drops of 18.41%, 18.10%, and 17.21%, respectively. The robustness of SpectraBayes can be attributed to its explicit low-pass denoising and probabilistic modeling mechanisms, effectively separating the high-frequency noise and handling the data uncertainty.

2) *Model Robustness Under Missing Value:* In this experiment, missing values are introduced by randomly masking individual time points with a predefined missing ratio, simulating the scenarios where sensor readings may be incomplete. We employ forward and backward fill to address this issue, ensuring data continuity and allowing models to process sequences without interruption. Model results are shown in Table VI. As the missing ratio increases from 0.05 to 0.25, the AUC-ROC decreases slightly from 79.24% to 78.06%,

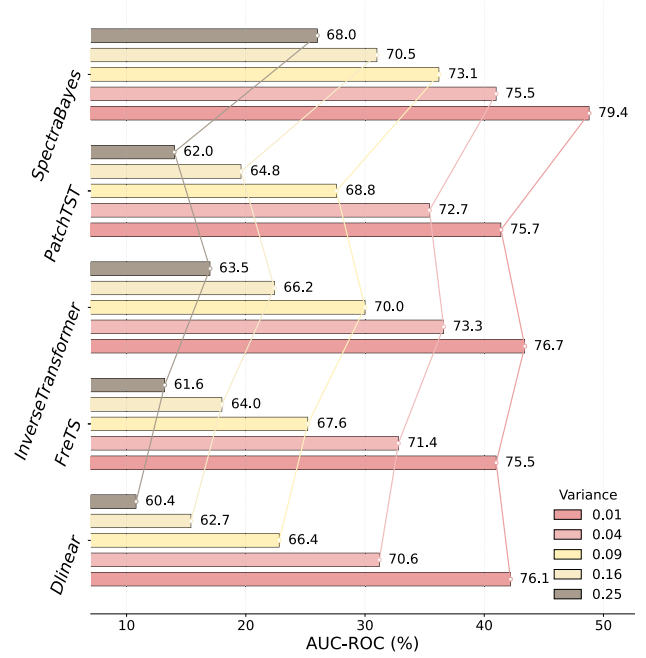


Fig. 7. Model performance (AUC-ROC) under varying noise levels (with Gaussian distribution variances set at different levels).

TABLE VI

MODEL DETECTION UNDER VARIOUS DATA MISSING RATIOS

Missing Ratio	AUC-ROC (%)
0.05	79.24
0.10	79.01
0.15	78.41
0.20	78.34
0.25	78.06

indicating the filling strategy is effective in moderate data missing.

G. Parameter Study

We tested how our model perform under different hyper-parameters setting, including the selection of Low-pass frequency F_p , the size of detection window T , and the prior distribution of Bayesian parameter.

1) *Setting of Low-Pass Frequency:* The effect of cutoff frequency F_p on the model performance was assessed across distinct datasets, as shown in Figure 8. In general, the performance of the model reveals a consistent and stable trend across the range of the F_p settings. For the 1-N and 1-W datasets, AUC-ROC performance showed almost no change across different low-pass frequencies (variation < 0.0002), with no discernible pattern. For the d090es0035 and d005es15531 datasets, AUC-ROC performance decreased slightly as the low-pass frequency increased.

2) *Detection Window and Stride Size:* Figure 9 summarize the AUC-ROC performance under different combinations of window size (ws) and stride size (ss). Across all datasets, window size exhibits a substantially stronger influence on detection performance than stride size.

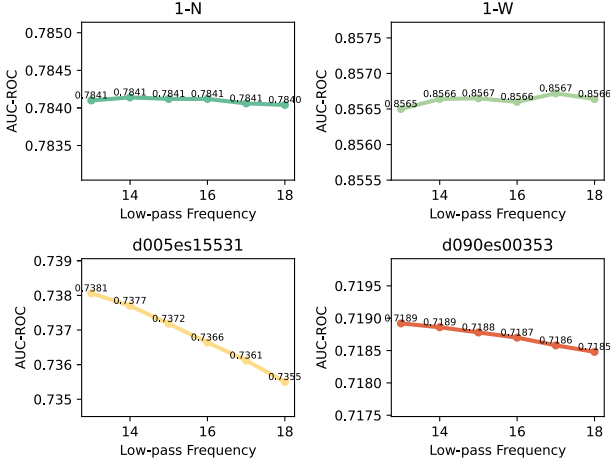


Fig. 8. Performance under different low-pass frequency on four selected datasets.

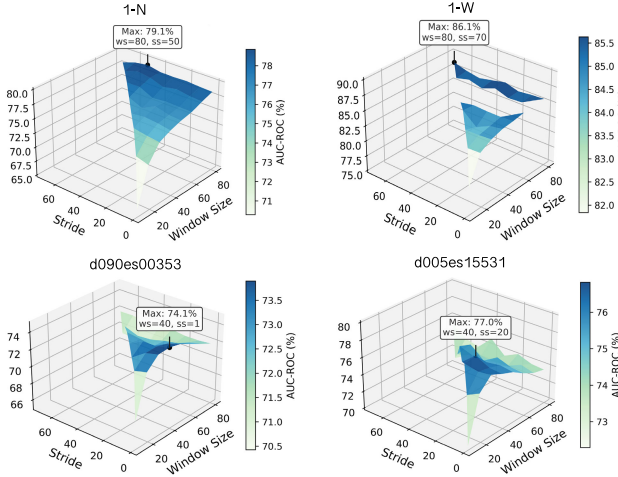


Fig. 9. Performance under different window and stride size on four selected datasets, ws and ss are abbreviation for window size and stride size, respectively.

For the Melbourne datasets (1-N and 1-W), AUC-ROC consistently improves as the window size increases, indicating that longer temporal contexts lead to more stable and discriminative representations for anomaly detection. This improvement is closely associated with an increase in the proportion of stationary sub-series under larger windows, suggesting that extended temporal aggregation enhances statistical stability in these datasets. In contrast, for the Seattle datasets (d005es15531 and d090es00353), performance peaks at moderate window sizes (typically $ws = 30-40$) and gradually degrades as the window size further increases. This behavior can be attributed to a decreasing level of stationarity under larger windows, where excessively long temporal contexts introduce stronger non-stationary variations that hinder anomaly discrimination.

Meanwhile, variations in stride size lead to only marginal performance changes when the stride is smaller than the window size. For a fixed window size, AUC-ROC remains relatively stable across a wide range of stride values, demonstrating that the proposed method is robust to stride selection.

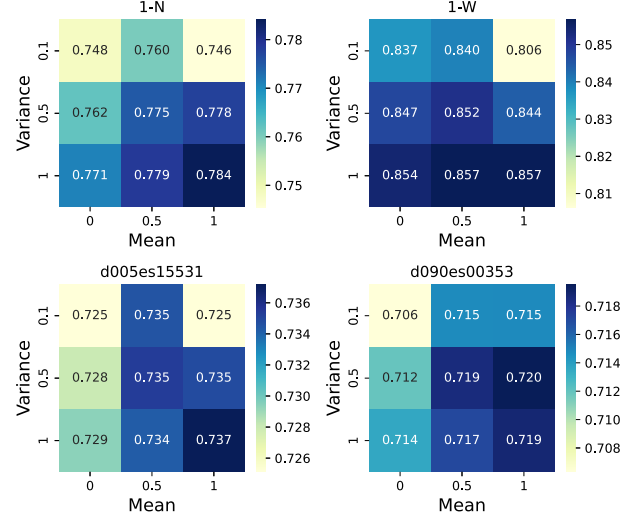


Fig. 10. Performance under different prior distribution setting on four selected datasets.

Overall, these results indicate that appropriate window size selection is critical for optimal performance.

3) *Setting of Prior Distribution:* We investigated the impact of prior distribution settings on model performance, as illustrated in Figure 10. The Gaussian prior setting governs the strength of the cross-spectrum learning, thereby influencing the correlation intensity learned between traffic series. A larger prior mean ($\mu_p = 1$) indicates a greater adjustment magnitude and a larger prior variance ($\sigma_p = 1$) indicates greater flexibility in correlation learning. Overall, as depicted in Figure 10, employing a higher prior mean ($\mu_p = 1$) and variance ($\sigma_p = 1$) generally leads to improved model performance. Specifically, prior variance demonstrates a particularly pronounced effect. Increasing it from 0.1 to 1 typically results in a significant performance enhancement (e.g., up to 3.8% on 1-N dataset). Prior mean also influences model performance, but to a lesser extent than prior variance. This observation suggests that macroscopic elements within the traffic system exhibit strong correlations and inherent uncertainty. Consequently, employing higher prior mean and variance values facilitates improved model performance.

V. CONCLUSION AND FUTURE WORK

This paper introduced SpectraBayes, which is a novel frequency-domain reconstruction model specially designed for traffic series anomaly detection. SpectraBayes improved the detection ability of frequency-domain model through the integration of periodicity embedding and probabilistic correlation learning. Extensive evaluations against a broad range of baselines demonstrated the superior performance of SpectraBayes on traffic series anomaly detection.

In response to the recent call by Sarteshnizi [1] for advancements in deep learning-based anomaly detection for traffic series anomaly detection, our comparative experiments involved a diverse array of deep learning models. Our findings indicated that architectural complexity alone is insufficient to ensure superior detection performance. Complex models

trained on noisy time series tend to overfit noise, diminishing the discrimination between normal and anomalous series. Instead, our results corroborate the importance of decoupled mechanisms, frequency representation, and correlation modeling as critical components for achieving effective traffic anomaly detection.

However, SpectraBayes remains a data-driven approach that lacks a certain level of physical interpretability, which holds significant potential for advancement. Combining the Fundamental Diagram with the Physics-Informed Neural Networks (PINNs) represents a potential solution. Explicitly integrating the domain-specific physical principles with machine learning techniques can enhance model interpretability in detecting anomalies. Moreover, the development of more comprehensive, system-wide datasets with detailed annotations is essential to support empirical studies, thereby fostering improvements in traffic system safety, efficiency, and resilience.

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Diyin Tang (Member, IEEE) received the B.S. and Ph.D. degrees from Beihang University, Beijing, China, in 2008 and 2015, respectively. From 2012 to 2013, she was a Visiting Ph.D. Student with the Department of Mechanical and Industrial Engineering, University of Toronto, Toronto, ON, Canada. She is currently an Associate Professor with the School of Automation Science and Electrical Engineering, Beihang University. Her research interests include fault prognostics, degradation-based modeling, and condition-based maintenance.



Fei Wang received the Ph.D. degree in computer architecture from the Institute of Computing Technology, Chinese Academy of Sciences, in 2017. From 2017 to 2020, he was a Research Assistant with the Institute of Technology, Chinese Academy of Sciences. Since 2020, he has been working as an Associate Professor with the Institute of Computing Technology, Chinese Academy of Sciences. His main research interests include spatiotemporal data mining, timeseries analysis, and AI for science.



Xi Zhu received the B.E. degree in electronic and information engineering and the M.E. degree in control science and engineering from Beijing University of Technology, Beijing, China, in 2010 and 2013, respectively, and the Ph.D. degree in signal and information processing from Beihang University, Beijing, in 2018. He is currently an Associate Researcher with the School of Electronic and Information Engineering, Beihang University. His research interests include multivariate time series analysis, spatiotemporal data mining, and target behavior cognition.



Ruoheng Li received the B.S. degree from Nanjing University of Aeronautics and Astronautics, Nanjing, China, in 2020, the B.S. degree from the Royal Melbourne Institute of Technology University, Melbourne, VIC, Australia, in 2020, and the M.S. degree from Nanjing University of Aeronautics and Astronautics in 2023. She is currently pursuing the Ph.D. degree with Beihang University. Her current research interests include data science and intelligent transportation.



Xianbin Cao (Senior Member, IEEE) received the B.E. and M.E. degrees in computer applications and information science from Anhui University, Hefei, China, in 1990 and 1993, respectively, and the Ph.D. degree in information science from the University of Science and Technology of China, Hefei, in 1996. He is currently a Professor with the School of Electronic and Information Engineering, Beihang University, Beijing, China. His current research interests include intelligent transportation systems, air traffic management, and intelligent computation.